Status	Finished
Started	Thursday, 3 September 2026, 1:47 PM
Completed	Thursday, 3 September 2026, 2:03 PM
Duration	15 mins 43 secs
Grade	**10.00** out of 15.00 (**66.67**%)

Question 1

Complete

Mark 1.00 out of 1.00

Why may an analog signal need to be modulated before transmission through a band-pass channel?

- a. To remove all frequency components from the original information
- b. To move the signal into a frequency range supported by the channel
- c. To convert a continuous signal directly into digital line coding
- d. To convert every analog value into a binary data element
- e. To reduce the carrier frequency permanently to zero hertz

The correct answer is: To move the signal into a frequency range supported by the channel

Question 2

Complete

Mark 1.00 out of 1.00

What is the fundamental idea of modulation?

- a. Maintaining all carrier characteristics unchanged during transmission
- b. Increasing all carrier characteristics by the same fixed amount
- c. Removing the carrier and transmitting only the original information
- d. Changing one characteristic of a carrier according to the information
- e. Converting every carrier signal into a sequence of binary numbers

The correct answer is: Changing one characteristic of a carrier according to the information

Question 3

Complete

Mark 3.00 out of 3.00

Considering the AM modulation, if the carrier frequency (fc) is 200 KHz, and the modulating signal has the frequency of 15 KHz,

- The total bandwidth required is 30 KHz.
- Hint (Slide 9): $BW_{(AM)} = 2*B_m$

Therefore, the lower frequency and the upper frequency of the AM modulated signal will be

- Lower frequency = 185 KHz.
- Upper frequency = 215 KHz.

Lower = 200 - 15 KHz

Upper = 200 + 15 KHz

Question 4

Complete

Mark 0.00 out of 2.00

Considering an audio signal with a bandwidth of **6 KHz**.

What is the bandwidth needed if we modulate the signal using **FM**?

Answer: 60 Hz

Hint (Slide 17): $BW_{(FM)} = 2*(1+b)B_m$ *and* *b = 4; Then* $BW_{(FM)} = 10\ B_m$

The correct answer is:
Considering an audio signal with a bandwidth of **6 KHz**.

What is the bandwidth needed if we modulate the signal using **FM**?

Answer: [60K] Hz

Hint (Slide 17): $BW_{(FM)} = 2*(1+b)B_m$ *and* *b = 4; Then* $BW_{(FM)} = 10\ B_m$

Question 5

Complete

Mark 5.00 out of 5.00

Consider an ASK signal with bandwidth spanning from 100KHz to 500KHz. Fill in the blanks.

- The bandwidth (diff between lower freq and upper freq) = 400 KHz
- The middle of the bandwidth = 300 KHz
- The carrier frequency = 300 KHz
- Define d = 1 and r = 1; BW = (1 + d) * S; BW = (1 + d) * (N/r)
- Hint: Slide 30,31,32

Therefore, the data rate (N) = 200 Kbps

And the baud rate (S) = 200 Kbaud/sec

The correct answer is:

Consider an ASK signal with bandwidth spanning from 100KHz to 500KHz. Fill in the blanks.

- The bandwidth (diff between lower freq and upper freq) = [400] KHz
- The middle of the bandwidth =[300] KHz
- The carrier frequency = [300] KHz
- Define d = 1 and r = 1; BW = (1 + d) * S; BW = (1 + d) * (N/r)
- Hint: Slide 30,31,32

Therefore, the data rate (N) = [200] Kbps

And the baud rate (S) = [200] Kbaud/sec

Question 6

Complete

Mark 0.00 out of 3.00

Consider an **ASK** signal with frequency from 2000 to 8000 Hz in **full-duplex** transmission mode. Fill in the blanks.

- The bandwidth for each direction is 6 KHz.
- Suppose the lower bandwidth is for forward transmission and another one backward transmission.

Therefore, the carrier frequency for forward is 6.5 KHz and the carrier frequency for backward is 4 KHz.

Hint: Slide 35

- The bandwidth for each direction is (8,000-2,000)/2 =3,000 Hz = 3 KHz.
- Suppose the lower bandwidth is for forward transmission and another one backward transmission.

Therefore, the carrier frequency for forward is 3.5 KHz and the carrier frequency for backward is 6.5 KHz.

fc1 fc2

2,000 **3,500** 5,000 **6,500** 8,000